Milestone 1 Submission Report

Formula 1 Podium Prediction | Introduction to Artificial Intelligence

Run Provenance: Kaggle notebook ACL_MS1, Version 7 of 7 (successful 113.1s run; 110 output files). Source revision 3f5da97; Formula 1 World Championship dataset (1950–2024). All figures and metrics extracted directly from Version 7.
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Kaggle Notebook: https://www.kaggle.com/code/crimewizard/acl-ms1
GitHub Repository: https://github.com/CrimeWizard/f1-podium-prediction

Executive Summary

This project builds an end-to-end, leakage-safe machine learning pipeline to predict whether a Formula 1 driver will finish on the podium (top-3 finish) based strictly on information available after Saturday qualifying and before Sunday race start. Using 14 relational tables from the Kaggle Ergast dataset (1950–2024), we conduct extensive data auditing, address historical shared-drive duplicates, and engineer pre-race championship standings and rolling form features. A strictly chronological train/validation/test split is enforced (train ≤ 2019, validation 2020–2021, held-out test 2022–2024). Given severe class imbalance (~13–15% podiums), PR-AUC is our primary evaluation metric. Among four model variants, regularized Random Forest attained the highest unseen test PR-AUC (0.6965), while a shallow Neural Network achieved the highest validation PR-AUC (0.7433). Both decisively outperformed a defaulted, overfitted tree baseline and linear logistic regression. All analytical conclusions are framed as descriptive historical associations rather than causal effects.

1. Task, Unit of Analysis, and Prediction Cutoff

The prediction unit is one driver-race entry uniquely identified by (raceId, driverId). The binary prediction target is 1 if the official finishing order satisfies positionOrder ≤ 3, and 0 otherwise. **Prediction Cutoff:** Information is locked after Saturday qualifying and prior to race start. Allowed features comprise penalty-adjusted grid slots, qualifying lap times, shifted championship standings, and prior-race historical form. Current-race results, finishing status, points scored, lap charts, pit stops, and updated standings are strictly excluded to eliminate target leakage.

2. Data Auditing, Cleaning, and Duplicate Handling

The relational schema (14 CSV tables) was ingested with immutable raw source integrity. Placeholder sentinels (literal \N) were parsed to nulls; dates and qualifying string durations were parsed to semantic datetimes and numeric seconds. **Key Integrity Audit:** Primary key audits across all 14 tables revealed zero duplicate and zero null keys. Foreign key reconciliations across 23 relational links showed zero orphan child records and zero broken references.
Shared-Drive Resolution: Early Formula 1 regulations (principally 1950s) permitted drivers to swap cars mid-race, producing multiple records per driver-race. In the complete historical dataset, 176 duplicate rows were identified (131 when excluding the non-championship Indianapolis 500 rounds). Our policy retains the single record with the best (lowest) positionOrder, capturing the driver's best verified finishing stint. The cleaned primary modeling table contains 26,287 unique driver-race records.

Exploratory Data Analysis (Before & After Cleaning)

EDA was conducted across both raw and cleaned pipeline stages. Figure 1 displays the raw distribution of finishing positions across all historical entries; Figure 2 illustrates the historical drift in qualifying row coverage per race, demonstrating that official session lap times were unrecorded prior to the mid-1990s and motivating our missingness indicators and imputation strategy. **Direct Impact of Data Cleaning (Figure 3):** Figure 3 directly evaluates the effect of cleaning on the data: (Left) A direct before-and-after comparison of positionOrder distributions contrasts raw entries against the cleaned modeling base, demonstrating the elimination of 176 shared-drive duplicate records (131 excluding non-championship Indy 500 rounds) to enforce a strict 1-to-1 driver-race grain. (Right) The annual podium rate post-cleaning confirms longitudinal target stability tightly bounded around ~15% across championship eras.

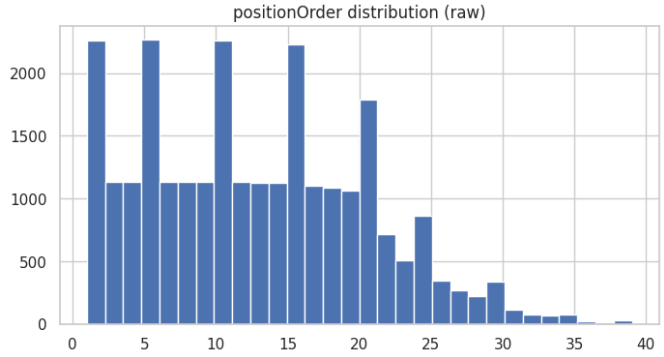


Figure 1: Distribution of raw finishing positions across uncleaned historical records.

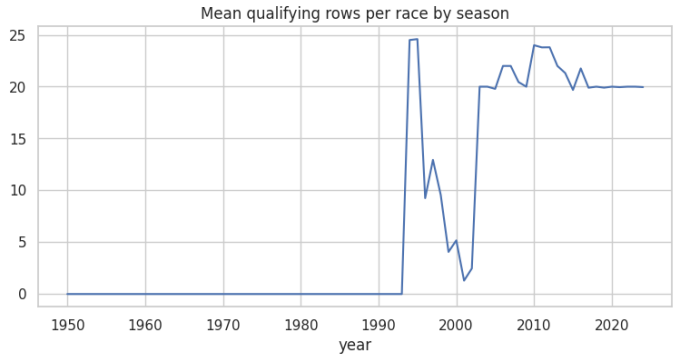


Figure 2: Historical qualifying data coverage drift: mean qualifying rows per Grand Prix by season.

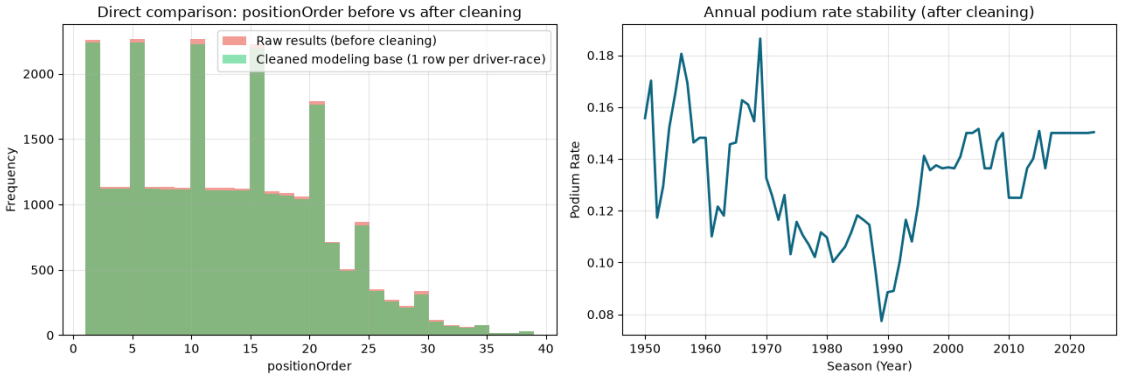


Figure 3: EDA before & after cleaning: (Left) Direct comparison of positionOrder before vs after deduplication; (Right) Cleaned annual podium rate stability.

3. Data-Engineering Questions (Multi-Hop Joins)

Q1. Front-Row Starts and Podium Conversion (Pre- vs Post-2014 Hybrid Era)

We distinguish official starting grid front rows (grid 1–2, post-penalties) from qualifying session front rows (quali 1–2). The multi-hop join combines results, races, circuits, and qualifying. Denominators exclude 1,601 non-starters and account for 14,231 legacy starters lacking electronic qualifying records. Circuits require ≥ 20 starts per era/basis combination. **Observed conversion rates:** Magny-Cours converted actual grid front rows highest pre-2014 (80.6%; 29/36); Yas Marina converted actual grid highest from 2014 (86.4%; 19/22). Suzuka converted qualifying front rows highest pre-2014 (76.9%; 20/26); Monaco converted qualifying highest from 2014 (85.0%; 17/20). Paired Spearman rank correlation across 7 common circuits is weakly negative: –0.180 for actual grid and –0.291 for qualifying. *Descriptive Note:* While conversion hierarchies shifted between eras, this correlation documents an empirical difference across time periods rather than proving a specific causal mechanism.

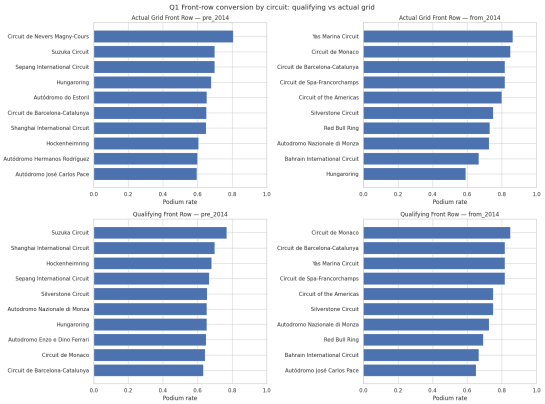


Figure 4: Front-row podium conversion rates by circuit, comparing pre-2014 vs from-2014 across actual grid and qualifying.

Q2. Home-Race Performance Controlling for Starting Grid Position

Home races are determined by matching normalized driver nationality to circuit host country (e.g., British → UK). To assess whether home drivers podium more often when starting from similar grid slots, entries are stratified into starting grid bins (1–5, 6–10, 11–15, 16+). Exclusions: 0 unknown-country records, 130 zero/missing grid starts. **Findings:** In Grid 1–5, home drivers achieved a 48.7% podium rate (181/372) vs 45.2% away (2,340/5,180; +3.5%). In Grid 6–10, home was 8.7% (36/416) vs 11.4% away (582/5,126; –2.7%). In Grid 11–15, home was 2.7% (11/414) vs 3.0% away (154/5,117). In Grid 16+, home was 0.3% (2/729) vs 0.7% away (53/7,202). *Descriptive Conclusion:* Controlling for grid reveals no consistent or systematic home advantage. Observed rates differ negligibly across bins and swing in opposite directions (+3.5% in top-5 vs –2.7% in midfield), showing that the observed rates show no consistent home advantage across the grid bins.

Grid Bin	Status	Starts	Podiums	Podium Rate
1–5	Away	5180	2340	0.4517
1–5	Home	372	181	0.4866
6–10	Away	5126	582	0.1135
6–10	Home	416	36	0.0865
11–15	Away	5117	154	0.0301
11–15	Home	414	11	0.0266
16+	Away	7202	53	0.0074
16+	Home	729	2	0.0027

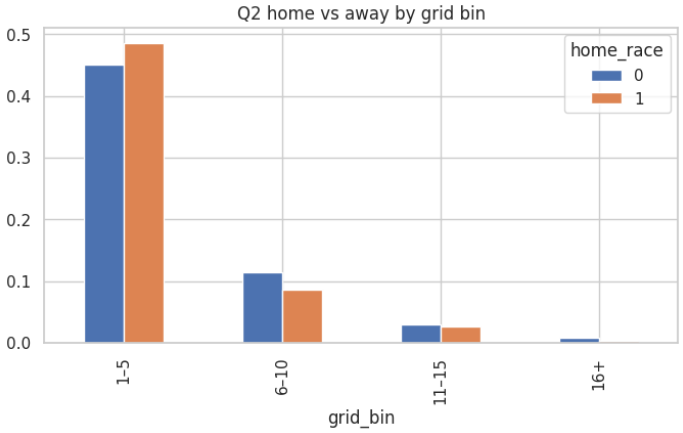


Figure 5: Home versus away podium conversion stratified by starting grid position bin.

Q3. Mechanical-Retirement Rates by Constructor (2014–2021 vs 2022–2024)

We categorize official status descriptions using keyword matching, isolating mechanical failures (power unit, turbo, electrical, hydraulic, gearbox, suspension) from collisions, driver errors, and regulatory disqualifications. Denominators include all race starters (1,601 non-starters excluded). **Descriptive Findings:** In 2014–2021, Caterham had the highest observed failure rate at 32.4% (11/34 starts). In 2022–2024, RB F1 Team had the highest observed rate at 10.4% (5/48 starts). Across constructors, observed mechanical retirement proportions were lower overall in 2022–2024 than in 2014–2021; we report this as a descriptive comparison across defined eras rather than attributing the trend to any single regulatory cause.

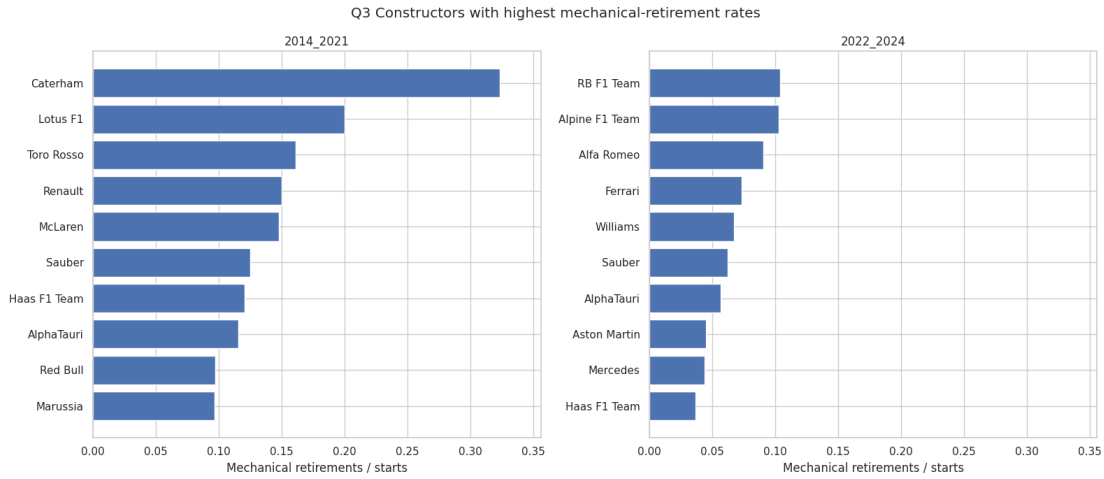


Figure 6: Mechanical retirement rates by constructor, comparing 2014–2021 vs 2022–2024 eras.

4. Leakage-Safe Feature Selection and Target Evidence

All features are strictly verified against the pre-race prediction cutoff. Championship standings are shifted to reflect driver and constructor standings *entering* the race weekend, resetting to zero at each season opener. Prior podium rate is computed over past career starts only.

Feature Group	Pre-Race Validity and Description
grid	Official starting grid slot after steward penalties; strongly predictive of race outcome.
quali_position, q1_sec, q2_sec, q3_sec	Saturday qualifying positions and lap times in seconds (available post-1994).
points_before, position_before	Driver championship points and rank entering the race (shifted, reset each season).
constructor_points_before, constructor_position_before	Constructor championship points and rank entering the race (shifted, reset each season).
podium_rate_prev	Cumulative rolling driver podium rate strictly over prior career races.

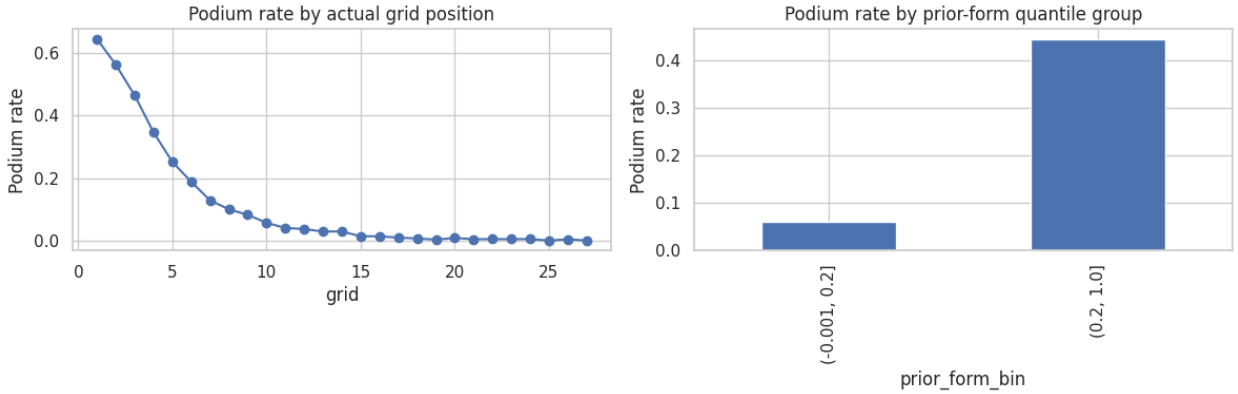


Figure 7: Empirical associations between pre-race features (grid slot and prior form) and podium probability.

5. Temporal Dataset Splitting and Preprocessing Experiments

To mirror actual deployment and prevent look-ahead bias, data is split chronologically: **Training:** 1950–2019 (24,148 rows; 12.58% podium prevalence); **Validation:** 2020–2021 (780 rows; 15.00% podium prevalence); **Held-out Test:** 2022–2024 (1,359 rows; 15.01% podium prevalence). All model hyperparameters and decision thresholds are selected strictly on training and validation splits.

Preprocessing Experiment	Validation PR-AUC
Impute then scale	0.7009
Scale then impute	0.7003
Median imputation	0.7009
Mean imputation	0.6906

Sensitivity Analysis: Excluding Indianapolis 500 rounds (1950–1960 non-championship technical regulations) produces identical validation PR-AUC and negligible change on test (<0.001 PR-AUC), confirming pipeline stability.

Indianapolis 500 Policy	Rows	Val PR-AUC	Test PR-AUC
exclude_indy_500 (Primary Policy)	26287	0.7009	0.6610
include_indy_500 (Sensitivity Check)	26668	0.7009	0.6620

6. Predictive Modeling and Evaluation

We evaluate four model configurations representing diverse mathematical approaches: (1) **Logistic Regression**: Linear log-odds baseline with L2 penalty, providing well-calibrated probabilities under linear assumptions. (2) **Default Random Forest**: Unconstrained bagged decision trees showing heavy overfitting. (3) **Regularized Random Forest**: Max-depth capped with min-samples-leaf constraints, preventing single-sample leaf memorization. (4) **Shallow Neural Network (FFNN)**: Feed-forward network with 1–2 dense hidden layers, ReLU activations, dropout regularization, and sigmoid output. **Primary Metric:** PR-AUC is chosen because podiums represent ~13–15% of samples; ROC-AUC is over-optimistic on rare classes. Thresholds were tuned on validation data (optimal threshold = 0.7584) and frozen before test evaluation.

Model	Split	PR-AUC	ROC-AUC	F1	Accuracy	Precision	Recall
Logistic regression	Train	0.5370	0.8914	0.5595	0.8797	0.5189	0.6070
Logistic regression	Validation	0.7009	0.9181	0.6877	0.8987	0.6397	0.7436
Logistic regression	Test	0.6610	0.9246	0.6667	0.8837	0.5852	0.7745
Random forest (default)	Train	0.9754	0.9976	0.9535	0.9878	0.9155	0.9947
Random forest (default)	Validation	0.6859	0.9128	0.6900	0.9090	0.7054	0.6752
Random forest (default)	Test	0.6480	0.9250	0.6410	0.8904	0.6303	0.6520
Random forest (regularized)	Train	0.6277	0.9242	0.5945	0.8967	0.5874	0.6017
Random forest (regularized)	Validation	0.7072	0.9237	0.7043	0.9128	0.7168	0.6923
Random forest (regularized)	Test	0.6965	0.9335	0.6420	0.8933	0.6468	0.6373
Shallow FFNN	Train	0.5799	0.9044	0.5725	0.8879	0.5501	0.5968
Shallow FFNN	Validation	0.7433	0.9093	0.7155	0.9154	0.7217	0.7094
Shallow FFNN	Test	0.6624	0.9264	0.6682	0.8926	0.6229	0.7206

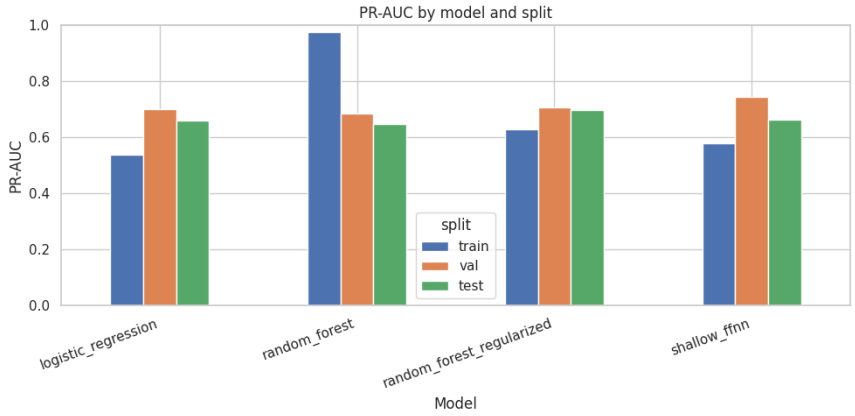


Figure 8: Comparative PR-AUC across models and chronological data splits.

7. Feature-Group Ablation Experiment

We systematically ablate one feature group at a time from the baseline model to quantify each component's marginal contribution. Figure 9 directly plots the held-out test PR-AUC across all feature groups alongside the full-feature baseline (0.6610, marked with a dashed red reference line). Removing **grid** triggers the steepest drop on unseen test data (Δ –0.0249 PR-AUC), confirming starting slot as the paramount predictor. Qualifying times (Δ –0.0101) and historical form (Δ –0.0056) provide necessary secondary refinements.

Feature Group Removed	Val PR-AUC	Held-Out Test PR-AUC
All Features (Baseline)	0.7009	0.6610
Drop Starting Grid	0.6749	0.6361
Drop Qualifying Session Times/Rank	0.6909	0.6509
Drop Championship Standings	0.7012	0.6565
Drop Rolling Driver Form	0.7071	0.6554

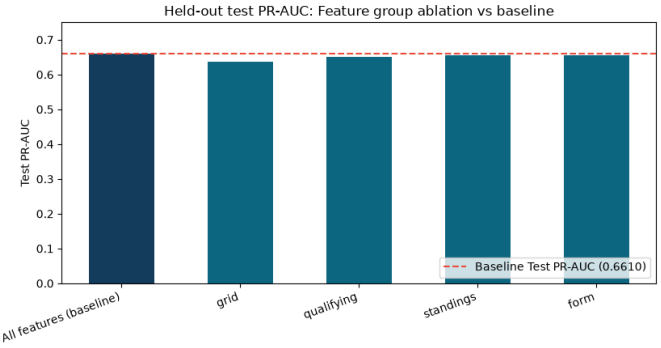


Figure 9: Impact of feature group ablation on held-out test PR-AUC, explicitly comparing ablated groups against the baseline bar and threshold (0.6610).

8. Model Explainability (XAI) and Detailed Interpretation

To inspect model decision rules globally and locally, we compute permutation feature importance, SHAP values, and LIME explanations. **Important Causal Disclaimer:** These explanations represent statistical associations learned by the model, *not causal effects*. A feature receiving high weight confirms model reliance, not that an external intervention on that feature would alter race outcomes.

Global Permutation Importance (Figure 10): Permuting starting grid degrades validation PR-AUC by ~0.15, confirming it as the primary anchor. Driver rolling podium form is second (~0.11), followed by constructor standings rank (~0.05).

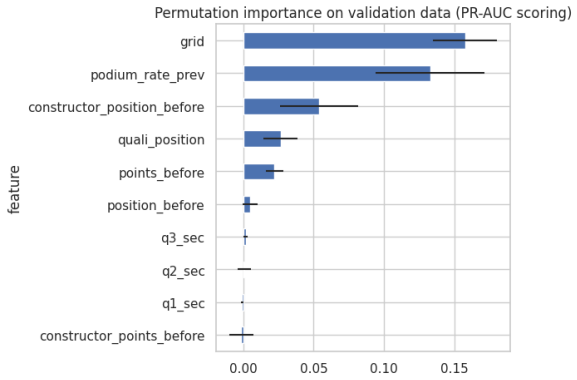


Figure 10: Permutation feature importance scored by PR-AUC drop on validation data.

Model Explainability Figures (Continued)

Global SHAP Beeswarm (Figure 11): Lower numerical grid values (front-of-grid starts) produce large positive SHAP contributions to podium log-odds, whereas starts outside the top 10 yield heavy negative penalties. High prior podium rate similarly shifts log-odds upward.

Local LIME Explanation (Figure 12): For an exemplary validation prediction (Lewis Hamilton, predicted podium probability 85.5%), LIME fits a local linear surrogate model whose feature weights indicate local importance. Favorable starting grid (num__grid ≤ –0.85) has a surrogate weight of +0.36 toward the podium class, and high historical form (num__podium_rate_prev > 0.33) has a surrogate weight of +0.16. These positive weights explain why the local surrogate favors the podium prediction, matching the primary model's 85.5% probability.

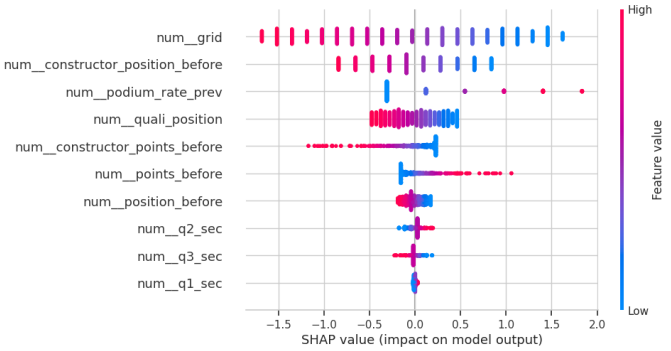


Figure 11: SHAP summary beeswarm plot illustrating feature value distributions and impact on model output.

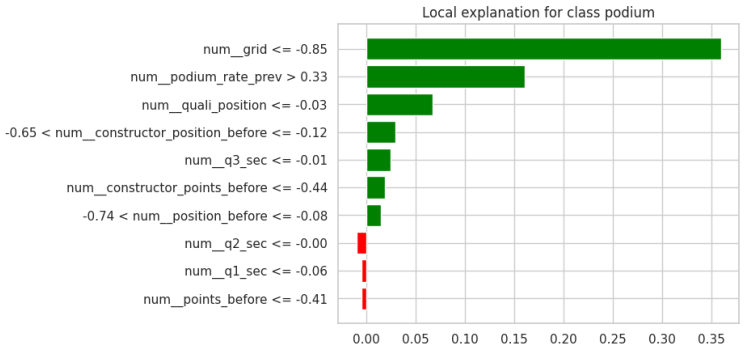


Figure 12: Local LIME explanation for an individual validation prediction (local surrogate weights for predicted prob: 85.5%).

9. Inference Function Demonstration

The production inference function (src/inference.py) processes single driver-race dictionary inputs, applying imputation and scaling pipelines consistently without data leakage. Tested on a complete validation record (Hamilton, grid 2), it predicted an **85.46%** podium probability (Class 1 at threshold 0.7584). Tested on a partially missing record (midfield entry lacking session qualifying lap times), the pipeline handled missing values via median imputation and returned a **40.55%** podium probability (Class 0), demonstrating production stability.

10. Limitations, Technical Constraints, and References

Key Limitations: (1) *Qualifying coverage drift:* Official lap times are absent prior to the mid-1990s, requiring imputation for early seasons. (2) *Constructor identity:* Frequent team acquisitions/rebrandings complicate historical constructor lineage. (3) *Sprint race formats:* Modern sprint weekends (introduced 2021) alter weekend tire allocation and starting grids, which are not separately modeled. (4) *Starter flag ambiguity:* Ergast lacks an independent starter flag; mechanical retirements before lights out are inferred from status labels.

References: Course Milestone 1 Companion Guide; Formula 1 for Newcomers Guide; Kaggle Formula 1 World Championship Dataset (Ergast); project repository and public notebook.